Retail360

Predicting 60-Day Repeat Purchase

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2,772 test customers | Test AP 0.755 | +2 buyers at 10% review capacity

LIVE APP

Abstract

Research question

Can past purchases predict whether a customer buys again within 60 days?

Why it matters

Retail analysts cannot inspect every customer. A useful ranking can help them focus limited review time on customers more likely to return.

What we built

We compare basic RFM with richer historical features, then show ranking quality, buyer counts at the same review capacity, score reliability, and original invoice evidence.

The analyst checks the evidence before acting.

Data Source

UCI Online Retail II

1,067,371 original transaction rows from 2009–2011. Source: Daqing Chen, UCI (DOI 10.24432/C5CG6D).

Fields and cleaning

Customer ID, invoice, product code, date, quantity, price, and country. We retain valid positive purchases and exclude cancellations.

Prediction protocol

Use each customer's prior 180 days; label a repeat purchase in the next 60 days. September 2011 test: 2,772 eligible customers.

Public app: 10 stored real cases for evidence review; no arbitrary live scoring.

Public historical data; no private records are uploaded.

Methodology

COHORT & TIME SPLIT

Features end at each cutoff; the following 60 days supply labels only. Train: Sep/Dec 2010 and Mar 2011 (n=9,659). Validate: Jun 2011 (n=2,659). Test: Sep 2011 (n=2,772).

180-day history → cutoff → 60-day outcome

FEATURES & MODELS

RFM uses recency, invoice count, and spending. Full history adds recent activity, invoice gaps, order value, and product variety. We compare an overall-rate dummy, logistic regression, and histogram gradient boosting.

Validation selected full-history logistic regression.

EVALUATION

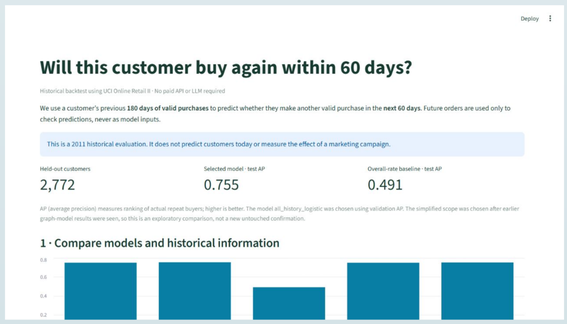
Choose by validation average precision (AP). On later cutoffs, compare AP, actual buyers found with equal review capacity, and score calibration. August/October checks overlap in customers.

Leakage check: no future order enters a feature.

SYSTEM ARCHITECTURE

UCI invoices → historical features → validated model → Streamlit score and invoice rows → human review.

WORKING PROTOTYPE



Public demo shows model comparison and 10 stored customer cases.

Results

Prediction quality

September 2011 test · average precision (AP)

Overall rate	0.491
RFM	0.750
Full history	**0.755**

Temporal sensitivity · Full − RFM AP

Aug	+0.0074 [95% paired CI: 0.0022, 0.0128]
Sep	+0.0049 [95% paired CI: 0.0006, 0.0094]
Oct	+0.0046 [95% paired CI: 0.0002, 0.0088]

AP gain over RFM: +0.005 in September.

Review-capacity sensitivity

September 2011 · same number reviewed

Top 5% · 139 review slots		
RFM 134		0
Full 134		
Top 10% · 278 review slots		
RFM 256		+2
Full 258		
Top 20% · 555 review slots		
RFM 458		-1
Full 457		

At 10%: only 2 more buyers among 278 reviewed.

At 20%: 1 fewer buyer. Aug/Oct at 10%: +1 each.

Probability reliability

Sep 2011 · G1 lowest score, G5 highest

	Raw	Adjusted	Observed
G1	18%	21%	26%
G2	22%	27%	35%
G3	29%	36%	43%
G4	42%	52%	60%
G5	71%	81%	82%

Brier: 0.222 → 0.209. All five groups still underpredict.

Richer features add only +0.005 AP beyond RFM. At a 10% review capacity, the full model finds two more actual buyers among 278 reviewed.

Conclusion

MAIN FINDINGS

Prior purchases rank repeat buyers: AP 0.755 versus 0.491 for the overall-rate baseline.

The improvement over RFM is small. At the top 10% review capacity, full history finds 258 actual buyers versus 256 for RFM. At 20%, it finds one fewer.

Even after validation-based adjustment, predicted rates remain below observed rates. Use scores for ranking and review, not exact personal odds.

HUMAN REVIEW

An analyst selects a stored customer, inspects historical features and source invoices, then records Accept, Reject, or Investigate. Future orders appear only for evaluation.

EVIDENCE PATH

Customer 14397 → 9 prior invoices → historical score 0.806 → invoice 563982, worksheet row 308453 → human review.

LIMITS & NEXT STEP

The focused scope followed earlier graph experiments. Later windows share customers. Test a new untouched period or retailer and improve calibration.

Acknowledgements

Thanks to

CS 5588 — Data Science Capstone
Instructor: Yugyung Lee

UCI Online Retail II is the public data source.

At a 10% review capacity, richer history found only two additional repeat buyers among 278 reviewed.